SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5
Duration : 60 Minutes

Total Marks: 50

Code of Conduct

I Rakshana AR (192126002 7) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. R

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

1) Compare Type 1 & Type 2 hypervisor for a cloud datacenter hosting mission critical applications. Analyze performance, security, and manageability, then justify the better choice.

1) Analysis of the situation:

A cloud datacenter hosting mission-critical applications must provide reliable performance, strong security, efficient resource utilization, & easy management. The choice of hypervisor directly affects these requirements because it determines how virtual machines access physical hardware.

2) Performance analysis:

A Type-1 hypervisor runs directly on physical hardware, whereas a Type-2 hypervisor runs on top of a host operating system.

In a Type-2 architecture, hardware requests pass through the host operating system and the hypervisor layer. This additional layer can introduce overhead & complete for resources with other host processor. Under heavy workloads, this may increase latency.

A Type-1 hypervisor has a more direct path to hardware resources and generally offers lower overhead. Therefore, it is better suited to workloads requiring consistent performance & high resource utilization.

3) Security analysis:

A Type-1 hypervisor avoids dependence on a general-purpose host operating system beneath the virtualization layer. This can reduce the attack surface, although its security still depends on configuration, Patching, & access controls.

A Type-2 hypervisor inherits risks from its host operating system. If the host is compromised, the virtual machines running on it may also be exposed.

Thus, a properly secured Type-1 architecture generally provides a stronger foundation for a production cloud datacenter.

4) Manageability analysis:

Type-1 hypervisors are commonly designed for centralized management, resource allocation, monitoring, and enterprise virtualization. These features help administrators balance workloads and manage large numbers of VMs.

Type-2 hypervisors are convenient for desktop testing & development but are generally less suitable as the primary architecture for large, mission-critical datacenters.

5) Justification & Conclusion:

A Type-1 hypervisor is the better choice because it generally offers lower virtualization overhead, a smaller underlying software stack, & more suitable enterprise-management capabilities. Products such as VMware ESXi are examples. However, high availability & security also require redundancy, monitoring, Patching, & appropriate access controls; the hypervisor alone cannot guarantee them.

2) A cloud provider must isolate workloads belonging to finance, healthcare, & student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

1) Analysis of the problem:

The cloud provider must run three different workloads on shared physical servers. Finance & healthcare applications may process highly sensitive information, while the student portal stores academic & personal records. If one workload can access another workload's data, it may cause privacy violations, data theft, or unauthorized modifications.

Therefore, the main challenge is to achieve efficient hardware sharing without compromising security.

2) How virtualization provides isolation:

a) Compute & memory isolation:

The hypervisor allocates CPU time & memory to individual VMs. Each VM operates within its own virtualized environment, limiting direct access to the memory & execution resources of other VMs. This reduces the risks of one workload interfering with another.

b) Network isolation:

Virtual networks, VLANs, and firewall rules can separate finance, healthcare, & student portal traffic. For example, a student portal should not be allowed to communicate directly with a finance database unless explicitly authorized.

c) Storage Isolation:

Each workload can use separate virtual disks, storage permissions, and encryption keys. Access-control policies ensure that an application can access only its authorized data.

Together, these mechanisms allow the provider to share physical hardware while maintaining logical separation between workloads.

3) Analysis of two major attack surfaces:

Attack surface 1: Hypervisor vulnerabilities:-

The hypervisor manages the boundary between VMs & physical hardware. If an attacker exploits a hypervisor vulnerability, they may attempt a VM escape, gaining access beyond the compromised VM. This could threaten other workloads hosted on the same server.

Attack surface 2: Virtual network vulnerabilities:-

Incorrect firewall rules, insecure virtual switches, or weak network segmentation may allow unauthorized traffic between VMs. For eg: a compromised student portal could attempt to communicate with a healthcare database if network access controls are misconfigured.

4) Recommended security design:

The provider should regularly patch the hypervisor, restrict administrative privileges, enforce network segementation, encrypt sensitive data, & monitor cross-VM communication. Separate security policies should be applied according to each workloads sensitivity.

3) Study the VM host utilization snapshot & recommend corrective actions: CPU 92%, Memory 88%, Storage I/O latency 35ms, & average VM boot time 170 seconds. Explain the likely bottlenecks.

1) Analysis of the given data:

Parameter	Observed Value	Analytical Interpretation
CPU utilisation	92%	Only 8% of CPU capacity is currently unused; Processing contention is possible.
Memory utilisation	88%	Only 12% of memory is unused; memory pressure may occur under additional demand.
Storage I/O latency	35ms	Storage requests may be experiencing delays; workload & storage type must be considered.
Average VM boot time	170s	Startup is slow enough to warrant investigation of resource contention & boot processes.

2) Identify the relationships between bottlenecks:

CPU & memory: when CPU utilisation reaches 92%, VMs may have to wait for additional workloads. If the host runs short of available memory, it may swap data to disk, increasing storage activity.

Memory & storage: If swapping occurs while storage latency is already elevated, the additional disk operations can further slow VM performance. However, high memory utilisation alone does not prove that swapping is occurring.

Storage & boot time: During startup, VMs read operating-system files, start services, & initialize applications. Slow storage responses can delay these operations. CPU contention or slow startup services may also contribute to the 170-second boot time.

3) Evaluate corrective actions:

First priority - Diagnose the bottleneck:

Examine CPU-ready time, memory swapping, disk queue length, I/O patterns, & VM boots logs. These measurements help distinguish CPU contention from memory pressure & storage delays.

Second Priority - Reduce resource contention:

If several VMs are overloading the same hosts, redistribute workloads or migrate suitable VMs to less-utilized hosts. This can release resources without immediately purchasing new hardware.

Third priority - Optimise the confirmed limited resource:

If memory swapping is significant, add memory or reduce excessive VM allocations. If storage is responsible, optimise I/O workloads or improve storage performance. Add CPU capacity if processing contention remains high.

Fourth priority - Verify the improvement:

Measure utilization & boot time again after making changes. This determines whether the chosen solution address the actual problem.

4) Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. support your answer with compute, storage, & network virtualization consideration:

1) Analysis of the problem:

A traditional datacenter often depends on physical servers, storage devices, & network equipment that require separate configuration & manual intervention. When business demand increases, administrators may need to install & configure additional hardware, which can delay service delivery.

2) Analysis of the three virtualization components:

a) Compute virtualization:

In a traditional datacenter, applications may be closely tied to specific physical servers. If a server becomes overloaded, redistributing workloads may require manual configuration.

With compute virtualization, a hypervisor runs multiple VMs on physical servers. Administrators can allocate CPU & memory resources, provision VMs, & move workloads between compatible hosts where supported.

b) Storage virtualization:

Traditional storage systems may require manual provisioning of individual storage devices or volumes. This can delay capacity expansion & complicate management.

Storage virtualization combines physical storage resources into logical pools. Administrators can allocate storage capacity to applications without managing every physical device separately.

c) Network virtualization:

Traditional networks often require administrators to configure physical switches, routers, & connections for each change. These changes can take time & may introduce configuration errors.

3) Overall comparison:

Factor	Traditional datacenter	SDDC
Resource Provisioning	Often manual	More automated
Scaling	May require physical changes	More flexible through software and pooled resources.
Management	Separate hardware-focused management	Centralized software-based management.
workload deployment	Can be slower	Generally faster.
Resource utilization	May leave resources underused	Virtualization can improve utilization.

4) Evaluation & Conclusion:

SDDC improves agility because compute, storage, & network resources can be provisioned & managed through software. For example, when an application experiences increased demand, administrators can allocate additional VM resources, provision storage, & adjust network policies more quickly than through purely manual hardware configuration.

5) A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity & Privacy design using federation concepts.

1) Analysis of the Problem:

A multi-cloud federation allows users & applications to access service across different cloud providers using trusted identity relationships.

Suppose a company uses one cloud provider for its main application & another for data storage. A user successfully authenticates with the first provider but is denied access to the second. This may happen because the providers interpret using identities, attributes, roles, or authentication tokens differently.

2) Analyze the causes & their effects:

a) Different identity formats:

One Provider may identify a user through an email address, while another uses a unique subject identifier. If these identities are not mapped correctly, the second provider may fail to recognize the user.

b) Different authorization policies:

A user may have the role of administrator in one cloud but a different role in another. If role mappings are inconsistent, the user may be denied legitimate access or granted excessive privileges.

c) Incompatible federation configuration:

The providers may use different authentication protocols, token formats, trusted issuers, or validation rules. Incorrect configuration can cause authentication failures even when the user is legitimate.

d) Privacy risks:

Sharing unnecessary personal information between providers increases exposure if a provider or account is compromised. Federation must therefore exchange only the information required for authentication and authorization.

3) Proposed secure federation design:

Step 1: Establish a trusted identity Provider.

Use a trusted identity Provider (IdP) to authenticate users & establish their identities. Participating cloud Providers act as relying Parties or service Providers that thrust the IdP under explicitly configured conditions.

Step 2: Use standard federation Protocols.

Implement Open ID Connect (OIDC) or Security Assertion Markup Language (SAML), according to the Providers supported capabilities. These standards enable identity information to be exchanged through defined mechanisms rather than incompatible custom implementations.

Step 3: Standardize identity & role mapping.

Define consistent user identifiers & map groups, attributes, & roles across providers. eg: a finance employee should receive only the finance-related permissions required in each cloud.

Step 4: Enforce secure authentication:

Validat token signatures, issuers, audiences, & expiration time. Apply role-based access control & least privilege. Use multi factor authentication for sensitive accounts & protect signing keys.

Step 5: Protect privacy & monitor access.

Exchange only necessary identity attributes, encrypt communication, & maintain audit logs. Regularly review access permissions & revoke access when users leave or change roles

4) Justification of the proposed design:

A trusted IdP reduces the need for users to maintain separate credentials for every provider. Standard protocols improve interoperability, while consistent role mapping addresses authorization mismatches. Token validation & least-privilege policies reduce unauthorized access, & data minimization limits unnecessary disclosure of personal information.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :